Kartikey Kesharwani

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Education

National Institute of Technology, Hamirpur **Aug 2024 – Aug 2028**
B.Tech in Electronics and Communication Engineering *Himachal Pradesh*
- CGPA: 8.5 / 10

Skills

Programming Languages: C, C++, Python, JavaScript, SQL
Electronics: Digital Electronics, Analog Electronics, Microcontrollers, Sensors & Instrumentation, Embedded Systems
Engineering: Signal Processing, Communication Systems, Control Systems, Troubleshooting & Debugging
Frameworks & Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, FastAPI, Flask
Tools & Platforms: Git, GitHub, Docker, CI/CD, Power BI, Excel, Arduino IDE, ESP32, MATLAB
Soft Skills: Leadership, Teamwork, Problem Solving

Experience

Data Analyst Intern **Jul 2026 – Aug 2026**
InAmigos *Remote*
- Collected, cleaned, and analyzed structured datasets using Python, SQL, and Excel.
- Developed interactive dashboards and reports to monitor key business metrics.
- Performed exploratory data analysis (EDA) to uncover trends, anomalies, and actionable insights.

Google Student Ambassador **Jul 2025 – Dec 2025**
Google Gemini *NIT Hamirpur*
- Represented Google Gemini as a Google Student Ambassador at NIT Hamirpur.
- Led product marketing initiatives that increased engagement and awareness for Google Gemini on campus.
- Acknowledged for leadership and initiative in promoting AI and tech engagement on campus.

Executive Member **Jan 2025 – Present**
Team Vibhav, Departmental Club of ECE *NIT Hamirpur*
- Led the Team RoboLimb project, achieving the **"Best Innovative Project Award"** for developing an AI robotic arm controlled by voice and vision.
- Organized multiple departmental and institutional events, engaging over 350 participants across all departments and **setting a record for participation** during Pre-Nimbus.

Projects

RoboLimb: ML Powered Robotic Arm | *YOLOv8, Vosk, OpenCV, Python, Arduino (C++)*
- Deployed an offline STT pipeline using Vosk for speech recognition with NLP-based command parsing, **achieving over 95% accuracy** without internet dependency, replacing invasive EMG/BCI interfaces for paralyzed and visually impaired users.
- Integrated a YOLOv8 real-time object detection module on a live webcam feed to spatially localize target objects, providing bounding-box coordinates directly to an inverse kinematics solver for precise 6-DOF arm control.
- Constructed a comprehensive voice → vision → motion pipeline: Vosk STT → NLP command parsing → YOLOv8 detection → position estimation → motion planning → servo actuation, enabling fully autonomous object grasping.

YouTube RAG | *LangChain, FAISS, Groq LLM, FastAPI, Streamlit*
- Built a production-style YouTube RAG chatbot using LangChain, FAISS, Groq LLM, FastAPI, and Streamlit to generate timestamped answers from YouTube transcripts.
- Implemented hybrid retrieval using FAISS + BM25, combining results with Reciprocal Rank Fusion (RRF) and Cross-Encoder reranking to retrieve the top 5 relevant transcript chunks.
- Integrated Ragas automated evaluation for Faithfulness, Context Recall, Context Precision, and Answer Relevancy, **achieving scores of 1.00, 1.00, 0.8875, and 0.8373** respectively.

Awards & Achievements

- **2nd Place, Hult Prize, NIT Hamirpur** – for developing RoboLimb, an AI-powered hardware solution integrating voice control and computer vision.
- **Solved 150+ DSA and SQL problems** on LeetCode and GeeksforGeeks – strengthened problem-solving skills and developed a strong understanding of data structures, algorithms, and database concepts.